

Sizing Calculation Spreadsheet

Vertical Geothermal Borefields

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Designers of vertical geothermal systems often need to quickly estimate the total length of a borefield for a given building. One way to perform this calculation is to use the sizing equation proposed by Kavanaugh and Rafferty¹ and contained in the *ASHRAE Handbook*.² This equation has been recast by Bernier³ into the following form:

$$L = \frac{q_h R_b + q_y R_{10y} + q_m R_{1m} + q_h R_{6h}}{T_m - (T_g + T_p)} \quad (1)$$

where

L is the total borehole length,

T_m is the mean fluid temperature in the borehole,

T_g is the undisturbed ground temperature,

T_p , the temperature penalty, represents a correction to the undisturbed ground temperature due to thermal interferences

between boreholes (in the case of a single borehole, $T_p = 0$),

q_y , q_m and q_h represent, respectively, the yearly average ground heat load (thermal annual imbalance), the highest monthly ground load and the peak hourly ground load,

R_{10y} , R_{1m} and R_{6h} are effective ground thermal resistances corresponding to 10 years, one month and six hours ground loads, and

R_b is the effective borehole thermal resistance.

Equation 1 was derived assuming that heat transfer in the ground occurs only by conduction and that moisture evaporation or underground water movement are not significant.

As illustrated in *Figure 1*, Equation 1 is based on the worst-case scenario represented by three successive thermal pulses with durations corresponding to 10 years, one month, and six hours. These pulse durations are typically used in design.

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The sizing calculation spreadsheet is available for free at www.ashrae.org/borehole.xls.

The method presented here is strictly applicable only when these values are used. The amplitudes of these pulses are determined from the building load profile and the coefficient of performance of the heat pumps. The evaluations of the three effective ground thermal resistances and of the temperature penalty are not straightforward. This article proposes correlations, based on multiple calculations, to calculate these values. Furthermore, a simple way to evaluate the equivalent borehole thermal resistance, based on the work of Hellström⁴ is proposed. With these correlations, a simple spreadsheet-based calculation can be carried out to obtain borehole length estimates.

In the next section, the methods used to generate the correlations are presented. This is followed by a brief description of the methodology used to evaluate R_b . Finally, two sizing examples, for single and multiple boreholes systems, are presented to demonstrate the ease of use of the proposed method. A Microsoft Excel spreadsheet accompanies this article (see link in the box above). The procedure presented here and the accompanying spreadsheet are not intended to replace commercially available borehole sizing software (which cover a larger spectrum of conditions) but to provide designers with a simple tool to guide them through ASHRAE Handbook calculations.

Correlations for R_{6h} , R_{1m} and R_{10y}

The effective ground thermal resistances account for transient heat transfer from the borehole wall to the far-field undisturbed ground temperature. Several ways exist to evaluate thermal resistances in the ground. In this work, the approach proposed by Kavanaugh and Rafferty¹ and contained in the ASHRAE Handbook² is used. It is based on the cylindrical heat source solution originally proposed by Carslaw and Jaeger⁵ used in conjunction with temporal superposition as proposed by Ingersoll and Plass⁶ and reviewed by Bernier.⁷ The effective thermal resistances are expressed as follows:

$$\begin{aligned} R_{6h} &= \frac{1}{k} G(\alpha t_{6h} / r_{bore}^2) \\ R_{1m} &= \frac{1}{k} [G(\alpha t_{1m+6h} / r_{bore}^2) - G(\alpha t_{6h} / r_{bore}^2)] \\ R_{10y} &= \frac{1}{k} [G(\alpha t_{10y+1m+6h} / r_{bore}^2) - G(\alpha t_{1m+6h} / r_{bore}^2)] \end{aligned} \quad (2)$$

where

G -function represents the cylindrical heat source solution,
 k is the ground thermal conductivity,
 α is the ground thermal diffusivity, and
 r_{bore} is the borehole radius (Figure 3).

The cylindrical heat source solution is strictly valid for one-dimensional (in the radial direction) transient heat transfer.

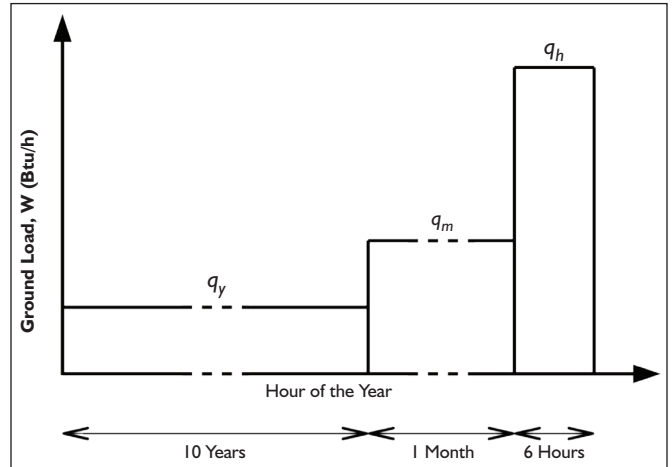


Figure 1: Three consecutive ground load pulses.

After a time period equivalent to $H^2/(90\alpha)$, where H is the borehole depth, Eskilson⁸ has shown that axial effects start to be significant. The error introduced when using the cylindrical heat source has been calculated by Philippe, et al.⁹

Based on these results, it appears that the axial effects are only significant for the R_{10y} term and that the error remains below 5% for typical values of thermal diffusivities. More accurate solutions, such as the two-dimensional finite line source model (Eskilson⁸) could be used. These solutions are more complex to solve and the gain in accuracy in the context of an engineering approximation does not warrant their use here. In the present work, the G -function is calculated precisely based on the work of Baudoin.¹⁰ Alternatively, readers can use the graphical values presented by Kavanaugh and Rafferty.¹ For each of the three effective ground resistances, R_{6h} , R_{1m} and R_{10y} , a total of 48 calculations are performed over the following range of typical operating conditions:

$$\begin{aligned} 0.05 \text{ m} &\leq r_{bore} \leq 0.1 \text{ m} \\ 0.025 \text{ m}^2 / \text{day} &\leq \alpha \leq 0.2 \text{ m}^2 / \text{day} \end{aligned} \quad (3)$$

Then, to avoid complicated calculations of the G -function, these results were curve-fitted in the following form:

$$\begin{aligned} R &= \frac{1}{k} f(\alpha, r_{bore}) \\ f &= a_0 + a_1 r_{bore} + a_2 r_{bore}^2 + a_3 \alpha + a_4 \alpha^2 + a_5 \ln(\alpha) + \\ &\quad a_6 \ln(\alpha)^2 + a_7 r_{bore} \alpha + a_8 r_{bore} \ln(\alpha) + a_9 \alpha \ln(\alpha) \end{aligned} \quad (4)$$

As shown, the resulting correlation function f depends only on two dimensional parameters, α and r_{bore} , given in m^2/day and in m , respectively. The correlation coefficients for f_{6h} , f_{1m} and f_{10y} are given in Table 1.

The correlated values are compared with the calculated values in Figure 2. The results are plotted in terms of effective thermal resistances for a particular value of ground thermal conductivity, $k=1 \text{ W}/(\text{m}\cdot\text{K})$ [$0.58 \text{ Btu}/(\text{h}\cdot\text{ft}\cdot^\circ\text{F})$]. The bottom and left axes present results in SI units while I-P units are used in the top and right

	f_{6h}	f_{1m}	f_{10y}
a_0	0.6619352	0.4132728	0.3057646
a_1	-4.815693	0.2912981	0.08987446
a_2	15.03571	0.07589286	-0.09151786
a_3	-0.09879421	0.1563978	-0.03872451
a_4	0.02917889	-0.2289355	0.1690853
a_5	0.1138498	-0.004927554	-0.02881681
a_6	0.005610933	-0.002694979	-0.002886584
a_7	0.7796329	-0.6380360	-0.1723169
a_8	-0.3243880	0.2950815	0.03112034
a_9	-0.01824101	0.1493320	-0.1188438

Table 1: Correlation coefficients for f_{6h} , f_{1m} and f_{10y} .

axes. As shown in this figure, the agreement between correlated and calculated values is very good with coefficients of correlation (R^2) equal to 99.99, 99.99 and 99.78, for f_{6h} , f_{1m} and f_{10y} , respectively. The values of f_{6h} and f_{1m} are much more sensitive to the characteristics of the soil and the borehole, while f_{10y} remains almost constant over typical ranges of values of α and r_{bore} .

Correlation for the Temperature Penalty, T_p

The temperature penalty, T_p , represents a correction applied to the ground temperature to account for the thermal interference between boreholes in a borefield. (T_p is actually a temperature difference, not an absolute temperature.) Bernier¹¹ has proposed a correlation to calculate T_p . This correlation is based on a correlation function, F , which depends on four parameters and takes the following form:

$$T_p = \frac{q_y}{2\pi kL} F(t/t_s, B/H, NB, A) \quad (5)$$

where

H is the borehole depth,

B is the distance between adjacent boreholes (a square mesh is assumed),

NB is the number of boreholes,

A is the geometrical aspect ratio (number of boreholes in the longest direction over the number of boreholes in the other direction), and

t_s is a characteristic time ($=H^2/9\alpha$).

The correlation function F is expressed as the sum of 37 terms with the following form:

$$F = \sum_{i=0}^{36} b_i \times c_i \quad (6)$$

with coefficients given in Table 2.

This correlation has some restrictions. It is valid for a constant value of B (i.e., the distance between adjacent boreholes, arranged in a square mesh, is the same throughout the bore-

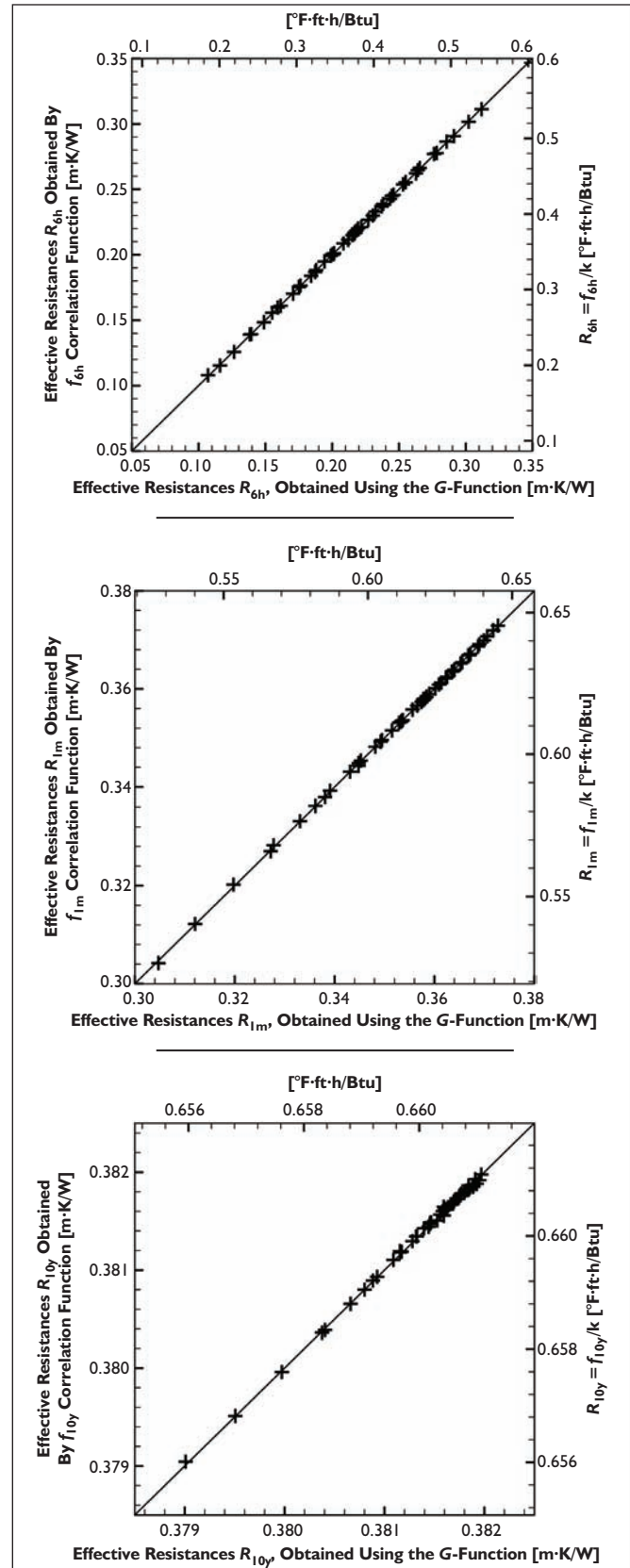


Figure 2: Comparison between the effective thermal resistances R_{6h} , R_{1m} , and R_{10y} obtained by calculation and correlation for $k=1$ W/(m·K) [0.578 Btu/h·ft·°F].

i	b_i	c_i
0	7.8189E+00	1
1	-6.4270E+01	B/H
2	1.5387E+02	(B/H) ²
3	-8.4809E+01	(B/H) ³
4	3.4610E+00	$\ln(t/t_s)$
5	-9.4753E-01	$(\ln[t/t_s])^2$
6	-6.0416E-02	$(\ln[t/t_s])^3$
7	1.5631E+00	NB
8	-8.9416E-03	NB ²
9	1.9061E-05	NB ³
10	-2.2890E+00	A
11	1.0187E-01	A ²
12	6.5690E-03	A ³
13	-4.0918E+01	(B/H) $\times$ $\ln(t/t_s)$
14	1.5557E+01	(B/H) $\times$ $(\ln[t/t_s])^2$
15	-1.9107E+01	(B/H) $\times$ NB
16	1.0529E-01	(B/H) $\times$ NB ²
17	2.5501E+01	(B/H) $\times$ A
18	-2.1177E+00	(B/H) $\times$ A ²
19	7.7529E+01	(B/H) ² $\times$ $\ln(t/t_s)$
20	-5.0454E+01	(B/H) ² $\times$ $(\ln[t/t_s])^2$
21	7.6352E+01	(B/H) ² $\times$ NB
22	-5.3719E-01	(B/H) ² $\times$ NB ²
23	-1.3200E+02	(B/H) ² $\times$ A
24	1.2878E+01	(B/H) ² $\times$ A ²
25	1.2697E-01	$\ln(t/t_s)$ $\times$ NB
26	-4.0284E-04	$\ln(t/t_s)$ $\times$ NB ²
27	-7.2065E-02	$\ln(t/t_s)$ $\times$ A
28	9.5184E-04	$\ln(t/t_s)$ $\times$ A ²
29	-2.4167E-02	$(\ln[t/t_s])^2$ $\times$ NB
30	9.6811E-05	$(\ln[t/t_s])^2$ $\times$ NB ²
31	2.8317E-02	$(\ln[t/t_s])^2$ $\times$ A
32	-1.0905E-03	$(\ln[t/t_s])^2$ $\times$ A ²
33	1.2207E-01	NB $\times$ A
34	-7.1050E-03	NB $\times$ A ²
35	-1.1129E-03	NB ² $\times$ A
36	-4.5566E-04	NB ² $\times$ A ²

Table 2: Coefficients b_i and c_i for the F correlation.

field). Furthermore, the other parameters are restricted to the following ranges:

$$\begin{aligned}
 -2 \leq \ln(t/t_s) \leq 3 \\
 4 \leq NB \leq 144 \\
 1 \leq A \leq 9 \\
 0.05 \leq B/H \leq 0.1
 \end{aligned} \quad (7)$$

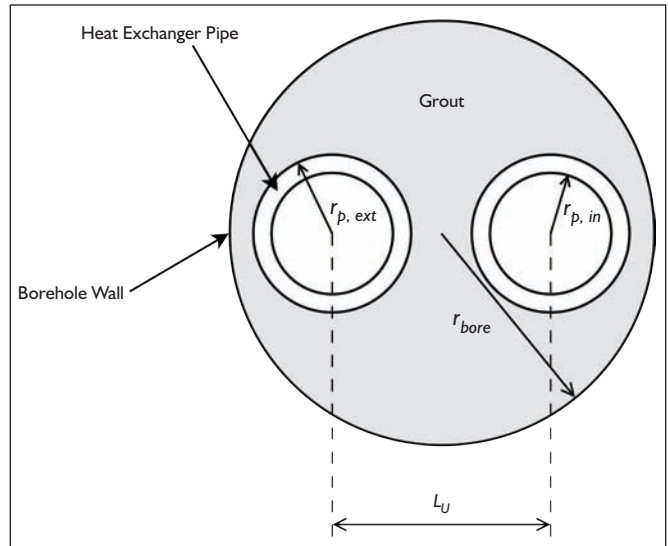


Figure 3: Cross-section of geothermal vertical borehole.

Bernier, et al.,¹¹ report that the difference in the T_p value when using Equation 6 is below 10% for most operating conditions compared to the standard g-functions of Eskilson.⁸ As shown in Equation 7, the correlation is valid for $NB \geq 4$. Borefields with two and three boreholes are not covered with the approach proposed here. For such cases, the temperature penalty must be calculated by the method proposed by Bernier¹¹ based on the work of Eskilson.⁸ However, borefields with two or three boreholes usually have small values of T_p as there is less borehole thermal interaction.

Calculation of R_b

A cross-section of a typical single U-tube geothermal borehole is shown in Figure 3. Typically, the borehole is filled with a grout to improve heat transfer and to avoid possible contamination between different aquifers. T_m , the average fluid temperature between the two legs of the U-tube, is assumed to be constant along the depth of the borehole and, therefore, equal to the average of the fluid inlet and outlet temperatures to the heat pumps ($T_m = (T_{in,HP} + T_{out,HP})/2$). The effective borehole thermal resistance R_b is the thermal resistance between the borehole wall and the fluid in the pipes (Figure 3). The effective borehole thermal resistance is based on three elementary effective thermal resistances and is given by:

$$R_b = R_g + \frac{R_p + R_{conv}}{2} \quad (8)$$

The three effective thermal resistances, R_{conv} , R_p and R_g , are, respectively, the convective resistance inside each tube, the conduction resistance for each tube and the grout resistance. They are obtained using the analytical equation proposed by Hellström.⁴ The three thermal resistances have the following expressions:

$$R_{conv} = \frac{1}{2\pi r_{p,in} h_{conv}} \quad (9)$$

$$R_p = \frac{\ln(r_{p,ext}/r_{p,in})}{2\pi k_{pipe}} \quad (10)$$

(11)

$$R_g = \frac{1}{4\pi k_{grout}} \left[\frac{k_{grout} - k}{k_{grout} + k} \ln \left(\frac{r_{bore}^4}{r_{bore}^4 - \left(\frac{L_U}{2}\right)^4} \right) + \ln \left(\frac{r_{bore}}{r_{p,ext}} \right) + \ln \left(\frac{r_{bore}}{L_U} \right) + \right]$$

where

h_{conv} is the film convection coefficient,
 $r_{p,in}$ and $r_{p,ext}$ are the inner and outer radii of the pipe,

k_{pipe} is the thermal conductivity of the pipe material,

k_{grout} is the thermal conductivity of the grout,

k is the ground thermal conductivity, and

L_U is the center-to-center distance between the two pipes (Figure 3).

Application Cases

Equations 1 to 11 have been implemented in an Excel spreadsheet. The calculation procedure for designing a single borehole or a borefield is illustrated in the flowchart presented in Figure 4. The spreadsheet can either be used for heating or cooling applications with proper signs for ground loads (positive ground loads correspond to heat rejection into the ground). The use of the spreadsheet will be illustrated using two examples for a single borehole and for a borefield. Results will be compared with those obtained from more sophisticated software tools and from data found in the literature.

As shown in Figure 5, there are four major parts in the spreadsheet: first set of inputs; first set of results; second set of inputs; and final results (which includes five sets of iterations). The iterations are required for multiple borehole configurations as T_p depends on H , which is not known *a priori*.

Single Borehole

The first application is for a single borehole in a cooling-dominated build-

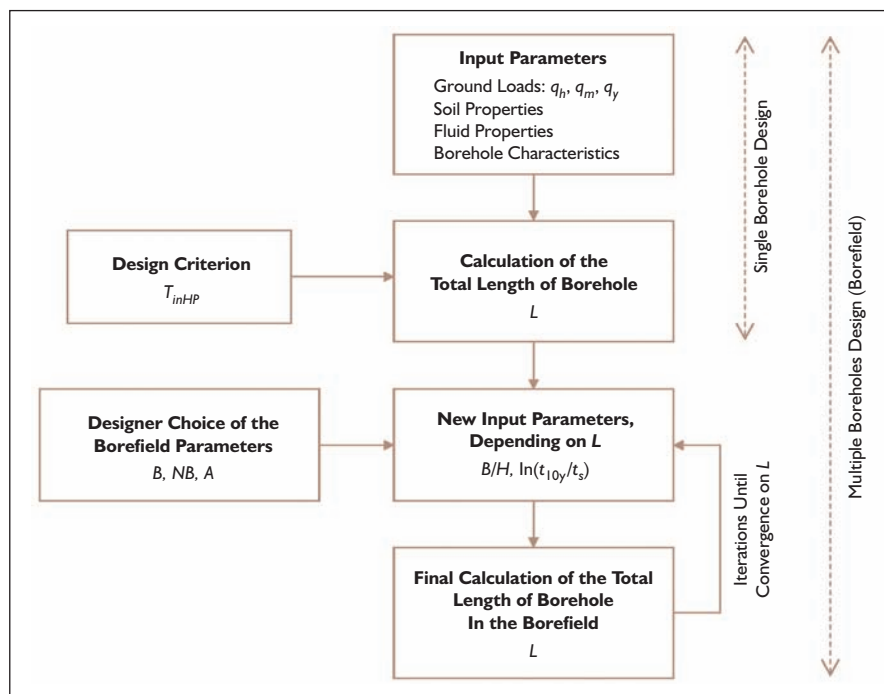


Figure 4: Flowchart of the calculation procedure for boreholes sizing.

ing. The three ground thermal pulses are 12 kW (40,000 Btu/h), 6 kW (20,000 Btu/h) and 1.5 kW (5,000 Btu/h), respectively. This is roughly equivalent to a 2.5-ton (8.8 kW) heat pump that rejects 12 kW into the ground at peak conditions. The monthly and yearly pulses can be estimated using hourly simulation results or equivalent full load operating hours.

Using this last method, it is estimated here that during the peak month, the heat pump operates half the time, so the monthly ground load is 6 kW. Finally, on an annual basis, the net amount of heat rejected into the ground is equivalent to a heat pump operating one-eighth of the time, which corresponds to 1.5 kW.

Then, ground properties and fluid thermal capacity are entered, as well as the total mass flow rate per kW of peak hourly ground load. The maximum (in cooling) or minimum (in heating) heat pump inlet temperature acceptable at peak conditions is entered next. This value, in fact, is the design criterion for sizing the borefield.

The next block of inputs concerns the borehole characteristics from the borehole radius to the internal film coefficient. The block for the first set of results

shows intermediate results on all effective thermal resistances, as well as the total length, which is 151.7 m (498 ft) in this case. The heat pump outlet temperature as well as the average fluid temperature in the borehole also are provided. They are obtained through an energy balance on the borehole. In the case of a single borehole, calculations stop here as there is no borehole thermal interference.

It is interesting to note the impact of a few key parameters. For example, in cooling mode, if the undisturbed ground temperature is 20°C (68°F) (instead of 15°C [59°F]), the length increases to 185.2 m (608 ft), a 22% increase. The last two values in the borehole characteristics block, L_U and h_{conv} have a relatively large impact on the effective borehole resistance (R_b) and, consequently, on the borehole length. For example, when the distance between the pipes is reduced to a point where the pipes are touching each other, i.e., L_u is reduced from 0.0511 to 0.0334 m (2 in. to 1.3 in.), the value of R_b increases from 0.120 to 0.143 m·K/W [0.246 °F·ft·h/Btu] (an 18% increase with a corresponding increase of the total length of 6.5% from 151.7 to 161.6 m [498 ft to 530 ft]). For turbulent flows, h_{conv} is usually above

1000 W/(m²·K) [176 Btu/h·ft²·°F] while for laminar flows it is generally below 100 W/(m²·K) [17.6 Btu/h·ft²·°F]. In the present case, everything else being equal, a laminar flow with $h_{conv} = 100$ W/(m²·K) [17.6 Btu/h·ft²·°F] would lead to a required borehole length of 174.5 m (573 ft).

Finally, the proposed approach was checked against the DST model,¹² which is often considered as a reference software tool for simulating ground heat exchangers. In this test, the DST model is run with three consecutive constant ground load pulses of 10 years, one month, and six hours using the data given in the first set of inputs. Results from the DST model give a total length of 150 m (492 ft), which is in good agreement with the value of 151.7 m (498 ft) obtained with the proposed approach.

Multiple Boreholes

In this second example, the data are provided by Shonder, et al.¹⁴ They are relative to a school and have been used to compare five different design programs against each other. This heating application uses a 12 × 10 borefield with 6.1 m (20 ft) spacing between boreholes. Much of the data in the first set of inputs is extracted from the comparison study¹⁴ except for the center-to-center distance between pipes that is assumed to be equal to 0.0471 m (1.85 in). This corresponds to a case where the distance between the pipes is the same as the distance between the pipes and the borehole wall. After analyzing test data achieved on various boreholes Remund¹³ recommended this spacing for such calculations.

For multiple boreholes, the procedure is a little more complicated than for single boreholes due to the presence of T_p in Equation 1. This temperature penalty depends on the borehole depth, which is the unknown *a priori*. An iterative procedure is required. The following three-step procedure is recommended to properly account for T_p .

First, calculations should be performed by assuming that T_p is zero as for a single borehole. This will lead to an approximate value of the total length of the

1st SET OF INPUTS			UNITS	Single borehole	Multiple boreholes	
Ground loads						
peak hourly ground load	q _h	W	12000	-392250		
monthly ground load	q _m	W	6000	-100000		
yearly average ground load	q _y	W	1500	-1762		
Ground properties						
thermal conductivity	k	W.m ⁻¹ .K ⁻¹	2	2.25		
thermal diffusivity	α	m ² .day ⁻¹	0.086	0.068		
Undisturbed ground temperature	T _g	°C	15	12.41		
Fluid properties						
thermal heat capacity	C _p	J.kg ⁻¹ .K ⁻¹	4200	4000		
total mass flow rate per kW of peak hourly ground load	m _{fhs}	kg.s ⁻¹ .kW ⁻¹	0.050	0.074		
max/min heat pump inlet temperature	T _{inHP}	°C	40.2	4.44		
Borehole characteristics						
borehole radius	r _{bore}	m	0.060	0.054		
pipe inner radius	r _{pin}	m	0.0137	0.0137		
pipe outer radius	r _{pext}	m	0.0167	0.0167		
grout thermal conductivity	k _{grout}	W.m ⁻¹ .K ⁻¹	1.50	1.73		
pipe thermal conductivity	k _{pipe}	W.m ⁻¹ .K ⁻¹	0.42	0.45		
center-to-center distance between pipes	L _U	m	0.0511	0.0471		
internal convection coefficient	h _{conv}	W.m ⁻² .K ⁻¹	1000	1000		
1st SET OF RESULTS						
Calculation of the effective borehole thermal resistance						
convective resistance	R _{conv}	m.K.W ⁻¹	0.012	0.012		
pipe resistance	R _p	m.K.W ⁻¹	0.076	0.071		
grout resistance	R _g	m.K.W ⁻¹	0.076	0.060		
effective borehole thermal resistance	R _b	m.K.W ⁻¹	0.120	0.102		
Calculation of the effective ground thermal resistances						
short term (6 hours pulse)	R _{sh}	m.K.W ⁻¹	0.114	0.101		
medium term (1 month pulse)	R _{1m}	m.K.W ⁻¹	0.180	0.160		
long term (10 years pulse)	R _{10y}	m.K.W ⁻¹	0.191	0.170		
Total length calculation assuming no borehole thermal interference						
heat pump outlet temperature	T _{outHP}	°C	45.0	1.1		
average fluid temperature in the borehole	T _m	°C	42.6	2.8		
total length	L	m	151.7	9899.3		
2nd SET OF INPUTS						
Borefield characteristics						
distance between boreholes	B	m		6.1		
number of boreholes	NB	-		120		
borefield aspect ratio	A	-		1.2		
FINAL RESULTS						
Total length calculation (with Tp)						
1st iteration						
distance-depth ratio	B/H	-		0.074		
logarithm of dimensionless time	ln(t _{10y} /t _s)	-		-1.120		
temperature penalty	T _p	°C		-0.240		
total borefield length	L	m		10151.5		
2nd iteration						
distance-depth ratio	B/H	-		0.072		
logarithm of dimensionless time	ln(t _{10y} /t _s)	-		-1.170		
temperature penalty	T _p	°C		-0.238		
total borefield length	L	m		10149.7		
3rd iteration						
distance-depth ratio	B/H	-		0.072		
logarithm of dimensionless time	ln(t _{10y} /t _s)	-		-1.170		
temperature penalty	T _p	°C		-0.238		
total borefield length	L	m		10149.7		
4th iteration						
distance-depth ratio	B/H	-		0.072		
logarithm of dimensionless time	ln(t _{10y} /t _s)	-		-1.170		
temperature penalty	T _p	°C		-0.238		
total borefield length	L	m		10149.7		
5th iteration						
distance-depth ratio	B/H	-		0.072		
logarithm of dimensionless time	ln(t _{10y} /t _s)	-		-1.170		
temperature penalty	T _p	°C		-0.238		
total borefield length	L	m		10149.7		
Final results						
total borefield length	L	m		10149.7		
borehole depth	H	m		84.6		

Figure 5: Spreadsheet for designing vertical geothermal boreholes—examples of results.

borefield. In the present example, the approximate length is 9899 m (32,470 ft).

Based on this approximate value, the designer enters the second set of inputs,

i.e., B (distance between the boreholes), NB (number of boreholes) and A (aspect ratio of the borefield). Depending on the available ground area and the ground

characteristics, the designer can opt for a more or less compact configuration. In the present case, $B=6.1$ m (20 ft), $NB=120$, $A=1.2$ which are the values chosen by the designer of the Maxey Elementary School.¹⁴ After this second set of inputs is entered, the final results block of the spreadsheet shows a set of five iterations in which T_p is reevaluated based on the new length calculations. This process converges rapidly and five iterations are usually sufficient. In the present case, the total borefield length is 10 150 m (33,290 ft) with a corresponding borehole depth of 84.6 m (277 ft). The temperature penalty T_p is -0.24°C (-0.43°F) after 10 years of operation. This is a relatively small value of borehole interference, which is due to the small annual thermal imbalance in the ground (1.762 kW). The design programs tested by Shonder, et al.,¹⁴ gave results between 65 m (213 ft) and 87 m (285 ft). Thus, the results given by the proposed procedure are in good agreement with other more sophisticated software tools.

Conclusion

A simple design procedure for single and multiple borehole configurations is presented. The procedure is based on the borehole sizing equation given in *ASHRAE Handbook*.² Simple algebraic correlations (Equation 4), based on a multitude of calculations, are proposed to calculate effective ground thermal resistances. These correlations, which are much simpler to use

than the cylindrical heat source solution, are shown to be in excellent agreement with calculated values. Borehole thermal interference is accounted for using a correlation developed by Bernier, et al.,¹¹ to evaluate the temperature penalty (Equation 6). Finally, the approach suggested by Hellström⁴ is used to evaluate the effective borehole resistance (Equation 8).

Designers using this method should be aware of its limitations. First, it is strictly valid for successive ground load pulse durations of 10 years, one month and six hours. Any significant deviations from these periods, especially the six-hour period, may require the use of commercially available borehole sizing software.¹⁵ Second, the range of applicable parameters for the determination of T_p is limited to values indicated in Equation 7. In the case of multiple boreholes, a simple iterative calculation is required as the temperature penalty depends on borehole depth, which is unknown *a priori*.

The proposed procedure is implemented in a spreadsheet. Two examples are provided, which show that the proposed approach is in good agreement with recognized borefield design software.

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